

Let's Get Started **With Intel Galileo**



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Intel Galileo is a part of Arduino family and is as simple to program even for non-programmers. Arduino philosophy is consistent in Intel Galileo as well. In this article, we shall start with Intel Galileo and then will be able to run our first program. Some more examples like glowing a single LED, multiple LEDs, RGB LED and reading a potentiometer are also presented at

the end to help you get hands-on with Galileo board. Intel Galileo board used in this article is shown in Fig. 1 and top view of the board is shown in Fig. 2.

Board interface

Following is the brief description of the components and interfaces of Galileo board:

5V power supply. A 2.1mm centre-positive jack for 5V regulated DC supply is used. Galileo board comes with AC-to-DC power adaptor having an output of 5V, 3A. Voltage input to Galileo board must not be more than 5V.

Ethernet interface. There is one 10Mbps/100Mbps Ethernet interface

on board. Ethernet control has a dedicated PHY.

RS232 interface. A 3-pin, 3.5mm jack (audio ring) forms the RS232 interface. Ring of the jack is transmitter (TX), tip is receiver (RX) and the sleeve is ground.

USB 2.0 client. To program Galileo board, this port is connected to the computer.

USB 2.0 host. USB2.0 full host supports USB device interface. Up to 128 devices can be connected to this port with the help of a USB device hub.

Arduino interface. There are two interfaces, namely, digital and analogue.

Digital interface. Galileo has 14 digital input/output pins. This in-

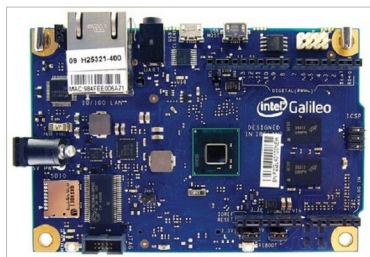


Fig. 1: Intel Galileo board

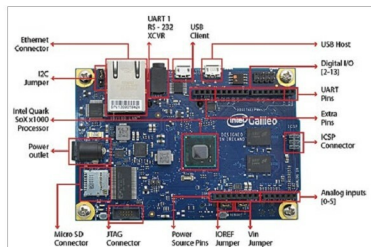


Fig. 2: Top view of Intel Galileo



Fig. 3: Bottom view of Intel Galileo with Intel Centriano N135 connected to PCIe

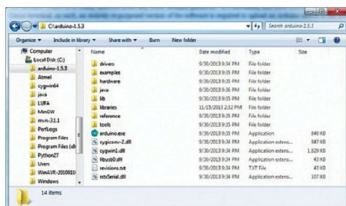


Fig. 4: Location of Arduino-1.5.3 folder